

INTERNSHIP WORK REGISTER

SGS FIELD TRACKER SYSTEM

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Register No.	THAYSCS026
Programme	BACHELOR OF COMPUTER SCIENCE HONOURS
Department	Department of Computer Science
Institution	St. Thomas College (Autonomous), Thrissur
Batch	2024 - 2028
Internship Organization	SGS / Italian Planters – Field Tracker System
Internship Status	ONGOING
Required Internship Hours	120 hours minimum

INTERNSHIP WORK REGISTER

Sl. No.	Date	Hours Worked	Cumulative Hours	Remarks
1	20 April 2026	2.0	2.0	Development / testing
2	24 April 2026	3.0	5.0	Development / testing
3	30 April 2026	3.5	8.5	Development / testing
4	7 May 2026	3.5	12.0	Development / testing
5	11 May 2026	4.0	16.0	Development / testing
6	18 May 2026	3.5	19.5	Development / testing
7	25 May 2026	3.5	23.0	Development / testing
8	5 June 2026	4.0	27.0	Development / testing
9	10 June 2026	3.5	30.5	Development / testing
10	17 June 2026	4.0	34.5	Development / testing
11	24 June 2026	4.0	38.5	Development / testing
12	27 June 2026	3.5	42.0	Development / testing
13	1 July 2026	4.0	46.0	Development / testing
14	2 July 2026	4.0	50.0	Development / testing
15	3 July 2026	2.0	52.0	Client review meeting
16	4 July 2026	4.0	56.0	Development / testing
17	7 July 2026	4.0	60.0	Development / testing
18	9 July 2026	4.0	64.0	Development / testing
19	11 July 2026	3.5	67.5	Development / testing
20	13 July 2026	4.0	71.5	Development / testing
21	15 July 2026	4.0	75.5	Development / testing
22	17 July 2026	2.0	77.5	Client review meeting
23	20 July 2026	4.0	81.5	Development / testing
24	22 July 2026	4.0	85.5	Development / testing
25	24 July 2026	4.0	89.5	Development / testing
26	27 July 2026	4.0	93.5	Development / testing
27	29 July 2026	4.0	97.5	Development / testing
28	31 July 2026	3.5	101.0	Development / testing
29	3 August 2026	4.0	105.0	Development / testing
30	5 August 2026	4.0	109.0	Development / testing
31	7 August 2026	4.0	113.0	Development / testing
32	10 August 2026	3.5	116.5	Development / testing
33	13 August 2026	4.0	120.5	Development / testing
34	14 August 2026	2.0	122.5	Development / testing

Sl. No.	Date	Hours Worked	Cumulative Hours	Remarks
35	16 August 2026	3.5	126.0	Development / testing
36	18 August 2026	4.0	130.0	Development / testing
37	20 August 2026	3.5	133.5	Development / testing

Total recorded hours in this draft: **133.5 hours**. The internship is ongoing.

DETAILED ACTIVITIES UNDERTAKEN

Date: 20 April 2026

Activities Undertaken

Participated in the initial online discussion regarding the proposed field-worker management system. Discussed the general problem, expected workflow, operational requirements, and the need for a system to manage field workers, worksites, assignments, attendance, and location-based operations. This provided the initial understanding of the project and its intended use in a service-sector environment.

Date: 24 April 2026

Activities Undertaken

Reviewed the initial project concept and studied the expected interaction between administrators, supervisors, field workers, worksites, and assigned jobs. Identified the major functional areas of the proposed system and began organizing the requirements into manageable modules for further development.

Date: 30 April 2026

Activities Undertaken

Worked on the initial technical investigation for location-based functionality. Studied Android location services, geofencing concepts, alarm triggering, and the practical requirements for detecting a worker entering or leaving a defined worksite boundary.

Date: 7 May 2026

Activities Undertaken

Reviewed the Italian Planters employee information received after the offline interaction with the client. Examined the employee list to understand the type of worker information that would be required in the Field Tracker system and considered how the information could be represented and managed within the application.

Date: 11 May 2026

Activities Undertaken

Developed and tested an Android proof-of-concept for geofencing and location-triggered alarms. The prototype was used to independently evaluate the location and alarm behaviour required for the proposed field-worker application. Testing also included handling cases where the Android location service was disabled and recovering the user interface when an existing alarm check continued running.

Date: 18 May 2026

Activities Undertaken

Continued testing of the Android geofencing and alarm prototype under different location and application states. Evaluated the feasibility of using the tested approach as a technical foundation for the location-triggering functionality required by the main Field Tracker application.

Date: 25 May 2026

Activities Undertaken

Studied the overall Field Tracker workflow and considered how the geofencing and alarm proof-of-concept could later be integrated into a larger worker application. Reviewed the distinction between independent technical testing and the complete application requirements.

Date: 5 June 2026

Activities Undertaken

Received and reviewed the latest SGS Field Tracker documents, including the System Overview, Proposed Features Document, and Architecture Diagram. Analysed the proposed workflow, modules, technical requirements, system interactions, and expected outputs. Identified areas requiring technical consideration before development and organized the requirements for subsequent planning.

Date: 10 June 2026

Activities Undertaken

Analysed the system architecture and major modules described in the project documentation. Studied the relationship between the web administration interface, worker application, backend services, database, location functionality, and operational workflows. Used the analysis to refine the development approach.

Date: 17 June 2026

Activities Undertaken

Worked on detailed requirement analysis and feature organization for the Field Tracker system. Considered worker management, worksite management, assignments, schedules, attendance, location tracking, and administrative functions and converted the requirements into a structured development plan.

Date: 24 June 2026

Activities Undertaken

Prepared and shared the proposed development roadmap and implementation plan for the SGS Field Tracker System. The roadmap covered project phases, delivery schedule, Web Admin Portal deliverables, Android Worker Application deliverables, backend milestones, infrastructure, deployment, and the proposed MVP release. Planned an iterative development approach so completed modules could be reviewed and validated progressively.

Date: 27 June 2026

Activities Undertaken

Worked on the technical planning of the web and mobile components based on the approved project direction. Considered the separation of administrative and worker-facing workflows and prepared the foundation for implementing the first development phase.

Date: 1 July 2026

Activities Undertaken

Finalized the initial mobile technology approach with Flutter for the worker application. Planned an Android-first implementation while keeping future iOS adaptation in consideration. Defined the Phase 1 foundation, including authentication, worker management, worksite management, assignments, registry views, worker login, home dashboard, assigned schedule, manual clock-in/clock-out, and worker profile.

Date: 2 July 2026

Activities Undertaken

Worked on the implementation and preparation of the Phase 1 system. Integrated the planned worker and administrative workflows and performed testing of the available functionality in preparation for the scheduled client review.

Date: 3 July 2026

Activities Undertaken

Attended the Phase 1 review meeting with the client at 11:00 AM. Demonstrated the Phase 1 Android application and provided the User Guide and Operations Manual, while also providing access to the web management portal. Recorded client feedback and the confirmed enhancements discussed during the review.

Date: 4 July 2026

Activities Undertaken

Analysed the additional requirements confirmed during the Phase 1 review. These included recurring job scheduling, bulk worksite import, leave handling, temporary worker reassignment with approval, additional worker information, job codes, public holidays, device-based login restrictions, configurable working duration, weekly offs, and database backup/restore.

Date: 7 July 2026

Activities Undertaken

Worked on the next development phase based on the confirmed requirements. Refined the worksite, worker, assignment, and scheduling workflows and considered how the additional administrative requirements would interact with the existing system.

Date: 9 July 2026

Activities Undertaken

Continued implementation and testing of the Field Tracker web and mobile components. Worked on data flow between the administrative functions and worker-facing schedules and examined the handling of worksite and assignment information.

Date: 11 July 2026

Activities Undertaken

Worked on backend and application integration and tested the system against the cloud-hosted development environment. Reviewed data synchronization and application behaviour while using the deployed backend and web environment for practical testing.

Date: 13 July 2026

Activities Undertaken

Continued development of the Field Tracker system with focus on worker schedules, assignments, and operational workflows. Tested the application using the current cloud-connected environment and identified issues requiring further refinement.

Date: 15 July 2026

Activities Undertaken

Investigated location-based functionality required for attendance and worksite tracking. Tested geofencing behaviour and examined the differences between the earlier standalone Android proof-of-concept and the integrated Flutter application.

Date: 17 July 2026

Activities Undertaken

Participated in the Phase 2 progress review meeting scheduled for 7:30 PM. Discussed the current implementation, development progress, technical issues, and the next steps required for the attendance, geofencing, tracking, and operational functionality.

Date: 20 July 2026

Activities Undertaken

Investigated difficulties encountered while implementing native-level location and alarm behaviour in the Flutter-based application. Compared the behaviour of the integrated application with the earlier standalone Android

prototype and identified areas where direct native Android control could be necessary.

Date: 22 July 2026

Activities Undertaken

Researched Flutter native packages and Android integration approaches for background location, geofencing, alarms, and related system-level behaviour. Evaluated possible approaches for improving reliability while maintaining the intended application architecture.

Date: 24 July 2026

Activities Undertaken

Investigated Kotlin and Kotlin Multiplatform approaches as possible alternatives for parts of the worker application requiring stronger native Android control. Compared the practical implications of retaining Flutter, using native packages, or moving towards a Kotlin-based implementation.

Date: 27 July 2026

Activities Undertaken

Performed application testing and troubleshooting with different Android versions and field-testing conditions. Investigated issues associated with changes in Android versions, device behaviour, location permissions, background execution, and the reliability of location-triggered functionality.

Date: 29 July 2026

Activities Undertaken

Worked with the cloud-hosted development environment for continued integration testing. The backend was deployed through the selected hosting environment and the web dashboard was deployed through the web hosting platform, allowing changes to be tested against the live development backend from the worker application.

Date: 31 July 2026

Activities Undertaken

Reviewed the growing scope of the web dashboard and worker requirements. Identified areas where the detailed requirements needed to be simplified or reorganized so that the core operational workflow could be implemented reliably without unnecessary complexity.

Date: 3 August 2026

Activities Undertaken

Continued technical research and development after the issues identified during field testing. Evaluated native Android control requirements and the feasibility of different technology approaches for reliable geofencing, alarms, background execution, and location services.

Date: 5 August 2026

Activities Undertaken

Worked on refining the system requirements and application workflow based on the technical issues encountered during development. Continued testing of the worker application and reviewed how the web dashboard should represent the essential operational information required by users.

Date: 7 August 2026

Activities Undertaken

Performed further testing, debugging, and technical evaluation of the worker application and backend-connected workflow. Documented implementation concerns and considered the most suitable approach for continuing

development while maintaining reliability and future platform compatibility.

Date: 10 August 2026

Activities Undertaken

Prepared for the next client review by consolidating the current implementation status, unresolved technical issues, and areas requiring clarification. Reviewed the requirements against the current system and identified information that needed confirmation before further development.

Date: 13 August 2026

Activities Undertaken

Reviewed the latest final project documents received from the client. Analysed the updated requirements covering inputs, outputs, site creation, scheduling, salary structure, and the overall system. Prepared a list of missing information, clarifications, technical concerns, and requirements requiring further discussion before implementation.

Date: 14 August 2026

Activities Undertaken

Participated in the client discussion regarding the updated project documents and requirements. Discussed the identified clarifications, technical concerns, system workflow, and the changes required before proceeding with the next stage of development.

Date: 16 August 2026

Activities Undertaken

Analysed the outcomes of the August requirement discussion and reviewed the impact of the updated requirements on the existing architecture and development approach. Began evaluating suitable technology, hosting, and implementation options for the revised system.

Date: 18 August 2026

Activities Undertaken

Prepared technical recommendations regarding the revised development approach, hosting options, and technology stack. Compared the practical requirements of the updated system with the limitations encountered during the earlier implementation and documented the recommended direction for client review.

Date: 20 August 2026

Activities Undertaken

Continued the ongoing development and documentation work following the latest requirement changes. Submitted the identified concerns, hosting options, and recommended technology stack for review. The project remains under development, with the updated requirements and supporting documentation currently being refined and implemented.

INTERNSHIP STATUS

The internship and the SGS Field Tracker System development are currently ongoing. The originally proposed project roadmap targeted the MVP/release milestone around 15 August 2026. Following subsequent client discussions and updated requirements, additional requirement clarification, technical evaluation, hosting evaluation, technology-stack review, implementation, and documentation are continuing. Therefore, this register records the work completed up to 20 August 2026 and does not represent the completion of the internship.

Signature of Internship Supervisor: _____

Name: _____

Designation: _____

Seal of Organization: _____